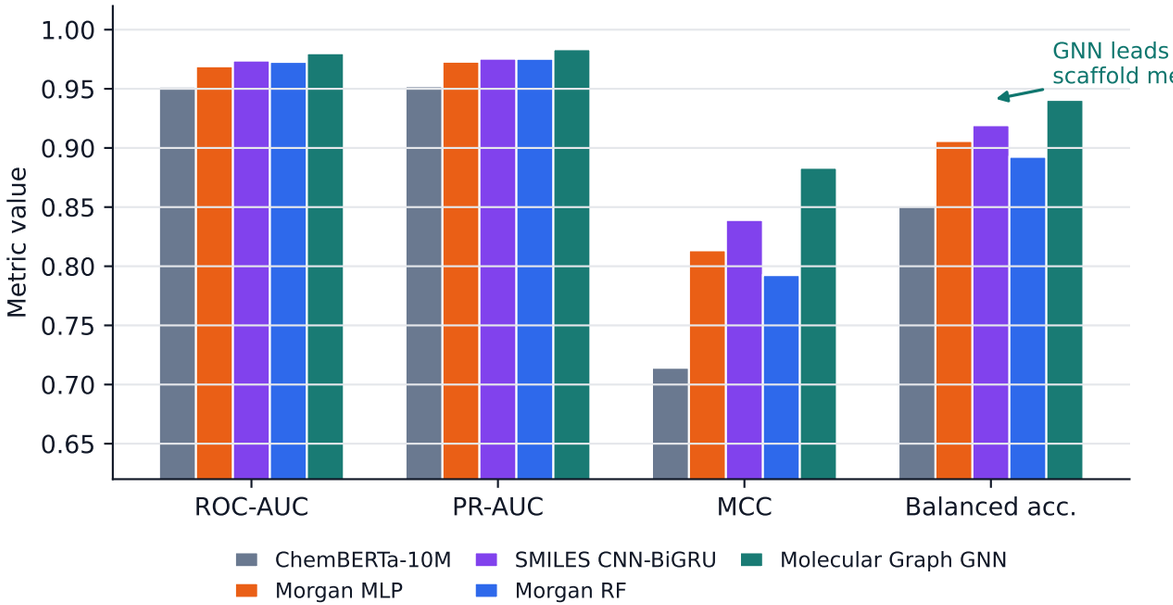
Structure-aware graph learning identifies the lead model for G4-ligand discovery

Key result: molecular graph GNN ranks first across all scaffold-split primary metrics and remains robust over repeated seeds.

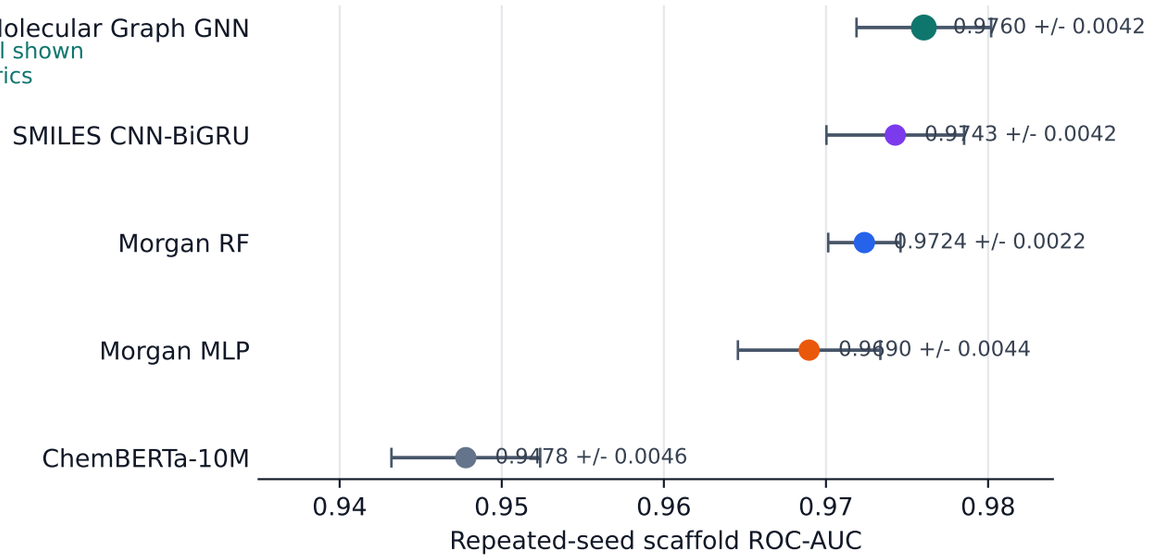
A

Scaffold split: GNN is the top model across ranking/classification metrics



B

Robustness over repeated scaffold splits



C

Manuscript-ready interpretation

Recommended claim: robust scaffold-aware ML prioritization, followed by molecular simulation support.
Avoid claiming true external validation until independent/prospective candidates are tested.

D

Key result for Q1 framing

A molecular graph GNN provides the strongest scaffold-level generalization, supporting its use as the primary model for external G4-ligand screening.

0.9801
Scaffold ROC-AUC

0.9835
Scaffold PR-AUC

0.8832
Scaffold MCC

1.00
Mean primary rank

Repeated scaffold ROC-AUC: 0.9760 +/- 0.0042